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GRID-FREE VORTEX METHODS FOR NATURAL CONVECTION IN TWO-DIMENSIONAL DOMAINS

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ABSTRACT

Vortex methods for simulating natural convection of an ideal gas in unbounded two-dimensional domains are presented. In particular, the redistribution method for diffusion is extended to enable simulation of nonlinear diffusion of an ideal gas in isobaric conditions encountered in unbounded low-Mach number flows. We also address the problem of handling source terms in grid-free vortex methods and propose a fast, accurate, and physically motivated method for solving the associated inverse problems. Examples include generation of baroclinic vorticity in non-reacting buoyancy driven flows, and in addition, generation of internal energy and species in buoyant reacting flows. Accuracy and speed of the proposed algorithms for nonlinear diffusion and vorticity generation are investigated separately. Simulations of natural convection of a “thermal patch” for Grashof number ranging from 1562.5 to 25000 are presented.

NOMENCLATURE

N Number of elements.
 h Desired elements spacing
 t Time.
 E Energy.

Δt Time step.
 k Thermal conductivity.
 c_p Constant pressure specific heat.
 p Pressure.
 R Ideal gas constant.
 T Temperature.
 L Characteristic length.
 $\vec{x}$ Position vector.
 $\vec{u}$ Velocity vector.
 $\vec{\omega}$ Vorticity vector.
 $\vec{f}_b$ Body force per unit mass.
 $\vec{g} \equiv -g\hat{z}$ Gravitational acceleration.
 ω Vorticity component normal to the flowfield.
 Γ Circulation.
 μ Dynamic viscosity.
 $\nu \equiv \mu/\rho$ Kinematic viscosity.
 π Dynamic pressure component.
 ρ Density
 $\alpha \equiv k/\rho c_p$ Thermal diffusivity.
 ϕ Core function.
 σ Core radius.
 Ma Mach number.
 Re Reynolds number.
 Pr Prandtl number.

Gr Grashof number.

Subscripts

∞ Free-stream conditions.

exp Volumetric expansion.

ω Vorticity related.

E Thermal energy related.

0 Conditions at beginning of time step.

INTRODUCTION

Vortex methods [1–3] are Lagrangian-based grid-free numerical methods for solving the vorticity transport equation, where the vorticity field is discretized using vortex elements. Operator splitting of the vorticity equation enables convection and diffusion of vorticity to be numerically carried out as separate steps. Vortex methods are adaptive in nature in the sense that computational elements carrying vorticity, moving with the flow field, exist only in regions of non-zero vorticity. Also the Lagrangian nature of the method eliminates numerical discretization of the convective term in the Navier-Stokes equations, thus eliminating associated numerical diffusion. Satisfactory handling of the no-slip boundary conditions at solid boundaries enabled high resolution spatially adaptive vortex methods [4–7] to simulate separating flows and accurately predict onset of separation and time evolution of drag and lift. The computational cost and accuracy of vortex methods are continuously improving. Replacing direct methods by fast methods [8] for velocity calculations reduced the cost from $O(N^2)$ to $O(N)$ or $O(N \log N)$. Diffusion schemes such as the Particle Strength Exchange Method [9] and the Redistribution Method [7, 10, 11] perform accurate simulation of the diffusion step at a cost that is a fraction of the convection step.

Grid-free Vortex Methods for simulating buoyancy driven and reacting flows still face various challenges that have been treated with different degrees of success. The first challenge is grid-free modeling of non-linear diffusion, where nonlinearity primarily arises from the fact that density is a function of temperature and pressure. A novel extension of the Smoothed Redistribution Method for simulating nonlinear diffusion is presented in this paper. Also, accurate handling of baroclinic vorticity generation, volumetric expansion, and the reaction terms in the energy and species equations in grid-free vortex methods continue to be a challenge. This is because the underlying problem is

an inverse problem of recovering the transported elements strengths (circulation, energy, etc.) from pointwise values of vorticity and temperature. To solve the inverse problem, Marshall and Grant [12] proposed a robust and quick iterative scheme what was later employed in [11] for simulating low Mach number, diffusion-controlled combustion in an axisymmetric domain. While Marshall and Grant's method does not suffer from the non-physical oscillatory behavior of the numerical solution arising from the poor condition number of the coefficients matrix, it does not yield an optimal solution in terms of point-wise accuracy for a given cost (or lowest cost for given accuracy). In this paper we propose an alternative physically motivated scheme for solving the inverse scheme.

This paper is organized as follows. The governing equations and the low Mach number approximation are presented next, followed by a brief discussion of Vortex Methods and operator splitting. Development of the redistribution scheme for non-linear diffusion of an ideal gas at isobaric conditions is discussed and presented in the section that follows. The numerical algorithm is then presented. The results section discusses accuracy and speed of the proposed implementation of nonlinear diffusion and presents simulations of natural convection of a thermal patch for different values of Grashof number.

GOVERNING EQUATIONS

In the low Mach number combustion model [13], the pressure is decomposed into a dynamic component, π , and thermodynamic component, p_∞ , the ratio of which is $O(Ma^2)$. This enables decoupling the density and pressure so that the total pressure may be replaced everywhere by the thermodynamic pressure, except in the momentum equation. As such density variations in an ideal gas arise only from temperature variations according to $p_\infty = \rho RT$. The total density variation is then used in the mass conservation equation to determine the divergence of velocity field $\nabla \cdot \vec{u} = -\frac{1}{\rho} \frac{d\rho}{dt}$, which is solved to compute the irrotational velocity component; one of two components of the total velocity.

The vorticity transport equation in two-dimensional domains, assuming constant μ , may be expressed as

$$\frac{d\vec{\omega}}{dt} = \nu \nabla^2 \vec{\omega} - \frac{\nabla T}{T} \times \left(\vec{f}_b - \frac{d\vec{u}}{dt} \right) - \frac{\vec{\omega}}{T} \frac{dT}{dt} \quad (1)$$

where $p_\infty = \rho RT$ has been employed. The energy equation, in absence of source terms, is

$$\frac{dT}{dt} = \alpha \nabla^2 T \quad (2)$$

where $\alpha \equiv \frac{k}{\rho c_p}$ and $d()/dt = \partial()/\partial t + \vec{u} \cdot \nabla()$. The vorticity transport equation and the energy equation may then be expressed in dimensionless form as

$$\frac{d\vec{\omega}}{dt} = \frac{1+T}{\text{Re}} \nabla^2 \vec{\omega} + \frac{\text{Gr}}{\text{Re}^2} \frac{1}{1+T} \left(\nabla T \times \left(\hat{z} + \frac{d\vec{u}}{dt} \right) - \vec{\omega} \frac{dT}{dt} \right) \quad (3)$$

$$\frac{dT}{dt} = \frac{1}{\text{RePr}} (1+T) \nabla^2 T \quad (4)$$

where velocity is nondimensionalized by $\sqrt{gL}$, time by $L/\sqrt{gL}$, temperature as $\frac{T-T_\infty}{T_\infty}$, where T_∞ is temperature at infinity, and density by $\rho_\infty = \frac{p_\infty}{RT_\infty}$. The dimensionless groups showing in the equations are

$$\text{Gr} = \frac{gL^3}{\nu_\infty^2}, \quad \text{Re} = \frac{\rho_\infty VL}{\mu}, \quad \text{Pr} = \frac{\mu c_p}{k} \quad (5)$$

VORTEX METHODS AND OPERATOR SPLITTING

In vortex methods, the vorticity, temperature, and species concentrations are expressed as

$$b(\vec{x}) = \sum_{i=1}^N S_i \phi_{\sigma_i}(\vec{x}, \vec{x}_i) \quad (6)$$

where S_i , the strength of element i , corresponds to circulation Γ when b is vorticity ω , to thermal energy E when b is temperature T , etc. The basis or cutoff function ϕ associated with element i is characterized by a cutoff or core radius σ_i . Choice of σ is based on two consideration: (i) accurate convection requires small elements overlap $\sigma_i/\bar{h}_i$, where $\bar{h}_i$ is the average element spacing in the neighborhood of element i and (ii) smooth distribution of b and consequently noise-free gradients ∇b require large $\sigma/\bar{h}$. Numerical simulations [11] showed that optimal value of $\sigma/\bar{h}$

lies in the range $2 < \sigma/\bar{h} < 5$. Note that, due to convection, the desired elements interspacing, used as the injection radius, is generally less than the actual average elements spacing.

Vortex methods rely on operator splitting in which within a numerical time step Δt , conserved quantities are first convected along particles trajectories and then diffusion is accounted for by solving the diffusion equation. In the presence of vorticity generation, a generation substep must also be included. The advection, diffusion, and generation sub-steps are

$$\text{Advection: } \frac{d\Gamma_i}{dt} = 0, \quad \frac{dE_i}{dt} = 0 \quad (7)$$

$$\text{Diffusion: } \frac{\partial \omega}{\partial t} = \frac{1+T}{\text{Re}} \nabla^2 \omega, \quad \frac{\partial T}{\partial t} = \frac{1+T}{\text{RePr}} \nabla^2 T \quad (8)$$

$$\text{Generation: } \frac{\partial \omega}{\partial t} = \frac{\text{Gr}}{\text{Re}^2} \frac{1}{1+T} \left(\nabla T \times \left(\hat{z} + \frac{d\vec{u}}{dt} \right) - \vec{\omega} \frac{dT}{dt} \right) \cdot \hat{z} \quad (9)$$

In the absence of solid boundaries, the velocity may be expressed as the sum of a (divergence-free) rotational component due to vorticity distribution and a potential component which accounts for free stream velocity and volumetric expansion

$$\vec{u} = \vec{u}_\omega + \vec{u}_\infty + \vec{u}_{\text{exp}} \quad (10)$$

where from conservation of mass

$$\nabla \cdot \vec{u}_{\text{exp}} = -\frac{1}{\rho} \frac{d\rho}{dt} = \frac{1}{1+T} \frac{dT}{dt} \quad (11)$$

Expressing $\vec{u}_{\text{exp}} = \nabla \phi$, the potential ϕ satisfies the Poisson equation $\nabla^2 \phi = (1+T)^{-1} dT/dt$.

NONLINEAR DIFFUSION: ISOBARIC IDEAL GAS

In this section, we propose a new method for grid-free computational modeling of nonlinear diffusion of an ideal gas at constant pressure in the context of vortex methods. The Smoothed Redistribution Algorithm [7,11] is extended to numerically simulate nonlinear diffusion, governed by

$$\frac{\partial T}{\partial t} = \frac{1}{\text{RePr}} (1+T) \nabla^2 T \quad (12)$$

while keeping the same basis function for linear diffusion in unbounded domains,

$$\phi_{\sigma_E} = \frac{1}{\pi\sigma_E^2} e^{-\frac{|\vec{x}-\vec{x}_0|^2}{\sigma_E^2}} \quad (13)$$

with $\sigma_E = 2h$ and $h = \sqrt{6\alpha\Delta t}$. The algorithm is based on invoking the linear diffusion algorithm a multiple times of the diffusion step $\Delta t/(\text{RePr})$.

To develop the redistribution scheme to solve Eqn. (12) over a time step $t \rightarrow t + \Delta t$, Eqn. (12) is approximated in an explicit form as

$$\frac{\partial T}{\partial t} \simeq \frac{1}{\text{RePr}} (1 + T_0) \nabla^2 T \quad (14)$$

where subscript 0 refers to conditions at the beginning of the diffusion step, and $T_0(\vec{x})$ given by

$$T_0(\vec{x}) = \left(\sum_{j=1}^N E_j \phi_{\sigma_E}(\vec{x} - \vec{x}_j) \right)_0 \quad (15)$$

We consider diffusion of element i of energy $(E_i)_0$, position $(\vec{x}_i)_0$ and core radius $(\sigma_i)_0$. Diffusing this element by redistribution is essentially transferring fractions $f_k(E_i)_0$ ($k = 1 \dots M$) of the element energy to its M neighbors such that the following moments are satisfied

$$\int \frac{\partial T}{\partial t} d\Omega = \frac{1}{\text{RePr}} \int (1 + T_0) \nabla^2 T d\Omega \quad (16)$$

Approximating the time derivative as a first order finite difference, then

$$\begin{aligned} \int x^m y^n \left(\sum_{k=1}^M f_k \phi_{\sigma_k}(\vec{x} - \vec{x}_k) - \phi_{\sigma_i}(\vec{x} - \vec{x}_i) \right) d\Omega &= \frac{\Delta t}{\text{RePr}} \\ \int x^m y^n \left(1 + \sum_{j=1}^{N^{(0)}} E_j^{(0)} \phi_{\sigma_j}(\vec{x} - \vec{x}_j^{(0)}) \right) \nabla^2 \phi_{\sigma_i}(\vec{x} - \vec{x}_i^{(0)}) d\Omega & \end{aligned} \quad (17)$$

where the right hand side is evaluated at the beginning of the time step, as in explicit methods. For the choice of the

core function of Eqn. (13), moments up to $m + n = 2$ are given, for constant σ , as

$$\sum_{k=1}^M f_k - 1 = \frac{\Delta\tau}{\pi\sigma^2} \sum_{j=1}^N E_j (\kappa - 1) e^{-\kappa} \quad (18)$$

$$\sum_{k=1}^M f_k \Delta x_{ki} = \frac{\Delta\tau}{2\pi\sigma^2} \sum_{j=1}^N E_j \Delta x_{ji} \kappa e^{-\kappa} \quad (19)$$

$$\sum_{k=1}^M f_k \Delta y_{ki} = \frac{\Delta\tau}{2\pi\sigma^2} \sum_{j=1}^N E_j \Delta y_{ji} \kappa e^{-\kappa} \quad (20)$$

$$\begin{aligned} \sum_{k=1}^M f_k \left(\Delta x_{ki}^2 + \frac{1}{2} \right) - \frac{1}{2} = \\ 2\Delta\tau + \frac{\Delta\tau}{4\pi\sigma^2} \sum_{j=1}^N E_j (\kappa(1 + \Delta x_{ji}^2) + \Delta x_{ji}^2) e^{-\kappa} \end{aligned} \quad (21)$$

$$\begin{aligned} \sum_{k=1}^M f_k \Delta x_{ki} \Delta y_{ki} = \\ \frac{\Delta\tau}{4\pi\sigma^2} \sum_{j=1}^N E_j \Delta x_{ji} \Delta y_{ji} (1 + \kappa) e^{-\kappa} \end{aligned} \quad (22)$$

$$\begin{aligned} \sum_{k=1}^M f_k \left(\Delta y_{ki}^2 + \frac{1}{2} \right) - \frac{1}{2} = \\ 2\Delta\tau + \frac{\Delta\tau}{4\pi\sigma^2} \sum_{j=1}^N E_j (\kappa(1 + \Delta y_{ji}^2) + \Delta y_{ji}^2) e^{-\kappa} \end{aligned} \quad (23)$$

where $\tau = \alpha t$, all the coordinates x_i, y_i, x_j, y_j and $\sqrt{\tau}$ are non-dimensionalized by σ and $\kappa = |\vec{x}_i - \vec{x}_j|^2/2$.

The redistribution algorithm for the non-linear diffusion problem is as follows:

1. Since $(1 + T)/(\text{RePr})$ may be interpreted as a local diffusion coefficient, elements that have larger temperature diffuse more than those of lower temperature. More specifically, for a given time step Δt , each elements diffuses over a diffusion step of $(1 + T)\Delta t/(\text{RePr})$. The diffusion algorithm adopted in this work is based on diffusion of an element by redistribution over a fixed diffusion step, equal to $\Delta t/(\text{RePr})$. This implies that over time step Δt , element i is diffused over $(1 + T_i)$ substeps, where in each substep, diffusion takes place over $\Delta t/(\text{RePr})$. So at the beginning of a diffusion step, the elements are organized into different categories based on their temperature. Category k ($k > 2$) contains all those elements with temperatures in the range

$[k-1, k]$. Each category k is then diffused over k substeps of $\Delta t / (\text{RePr})$.

2. The right hand side of the redistribution equations, along with the temperature at elements positions, are calculated once at the beginning of each time step. Given that the core function is a Gaussian, the cost of this step can be reduced by introducing a cutoff radius R_c/σ (between 3 and 5).

The linear system governing fractions k is solved subject to the constraint $f_k \geq 0$ using Simplex Method for linear programming, as presented in [10], or non-negative least squares, as proposed in [7, 11].

NUMERICAL ALGORITHM

The numerical algorithm employed for simulating natural convection in unbounded domains, for a typical time step k from t_k to $t_k + \Delta t$, is as follows:

1. Compute the particles velocities using a second order Runge-Kutta predictor corrector scheme. The vortical and potential velocity components are computed using a fast $O(N \log(N))$ algorithm.
2. Update elements positions (Eqn. (7)) according to $\vec{x}_i^* = \vec{x}_i + \vec{u}_i \Delta t$, for $i = 1, N$
3. Solve the nonlinear diffusion problems for vorticity and temperature (Eqn. (8)) using the Redistribution scheme (Eqns. (18-23)).
4. $\vec{x}_i^{(k+1)} = \vec{x}_i$ and $E_i^{(k+1)} = E_i$ for $i = 1, N$.
5. Compute $\omega(\vec{x}_i)$, $T(\vec{x}_i)$ and $\theta(\vec{x}_i) = (1 + T(\vec{x}_i))^{-1} dT(\vec{x}_i)/dt$ at particles positions. Approximate $dT/dt = (T(\vec{x}_i) - T(\vec{x}_i)^{(k)})/\Delta t$ with $T(\vec{x}_i) = \sum_{i=1}^N E_i \phi_\sigma(\vec{x}, \vec{x}_i)$ and $T(\vec{x}_i)^{(k)} = \sum_{i=1}^N E_i^{(k)} \phi_\sigma(\vec{x}, \vec{x}_i^{(k)})$.
6. Find the vorticity generated $\Delta\omega(\vec{x}_i)$ at particles positions using Eqn. (9).
7. Find the generated circulations $\Delta\Gamma_i$ by solving the inverse problem $\sum_{i=1}^N \Delta\Gamma_i \phi_\sigma(\vec{x}, \vec{x}_i) = \Delta\omega(\vec{x}_i)$
8. Update the elements circulations $\Gamma_i^{(k+1)} = \Gamma_i + \Delta\Gamma_i$, for $i = 1, N$.
9. Determine the particles (volumetric expansion) strengths $S_i^{(k+1)}$ by solving the inverse problem $\sum_{i=1}^N S_i^{(k+1)} \phi_\sigma(\vec{x}, \vec{x}_i) = \theta(\vec{x}_i)$.
10. Store elements positions $\vec{x}_i^{(k+1)}$, circulations $\Gamma_i^{(k+1)}$, strengths $S_i^{(k+1)}$, and energies $E_i^{(k+1)}$.

Steps 7 and 9 involve solving the inverse problems for transforming pointwise vorticity generation and volumetric expansion into integral quantities. This is accomplished by satisfying Eqn (6) at the center of each element resulting in the linear system

$$b(\vec{x}_j) = \sum_{i=1}^N S_i \phi_{\sigma_i}(\vec{x}_j, \vec{x}_i), \quad j = 1 \dots N \quad (24)$$

which is numerically solved, using the Projected Restarted Iteration Method (PRIM) [14], subject to the physically motivated constraint $S_i f_i \geq 0$, for $i = 1 \dots N$.

RESULTS

In this section, accuracy and speed of the Smoothed Redistribution Method proposed for non-linear isobaric diffusion of an ideal gas are investigated for a diffusing hot patch. Also, simulations of buoyancy driven flow of a thermal patch in an unbounded domain are presented for different values of Grashof number.

Nonlinear Isobaric Diffusion of a Hot Patch

To validate the proposed redistribution method for non-linear isobaric diffusion of an ideal gas, diffusion of a thermal patch of initial temperature $T_0(\vec{x}) = \frac{E_0}{\pi\sigma_0^2} e^{-|\vec{x}|^2/\sigma_0^2}$ and $\sigma_0 \simeq 0.1$ is simulated. Radial temperature profiles for $\alpha \Delta t = 2.5 \times 10^{-5}$, 5×10^{-5} , 10^{-4} , and 2×10^{-4} are plotted in Fig. 1 at $t = 0.2$ and $t = 1$ for $E_0 = 0.2$, corresponding to maximum initial temperature $T_{0,max} = 5.3$.

Temperature distribution (T) predicted by the proposed scheme are compared to numerical solutions (T^*) obtained by converting the PDEs to ODEs using a second-order accurate spatial discretization with subsequent time integration of the resulting ODEs using Runge-Kutta or variable order schemes (depending on stiffness of the problem). The grid size is chosen to be $\Delta r = 0.005$ and time integration is carried out with 0.01% relative accuracy. The comparison presented in Fig. 1 shows that the proposed nonlinear diffusion scheme predicts well the time evolution of the temperature distribution in the patch in the presence of large temperature gradients similar to those encountered in combustion. The % L_2 relative error in the temperature integrated over a uniform grid, computed as

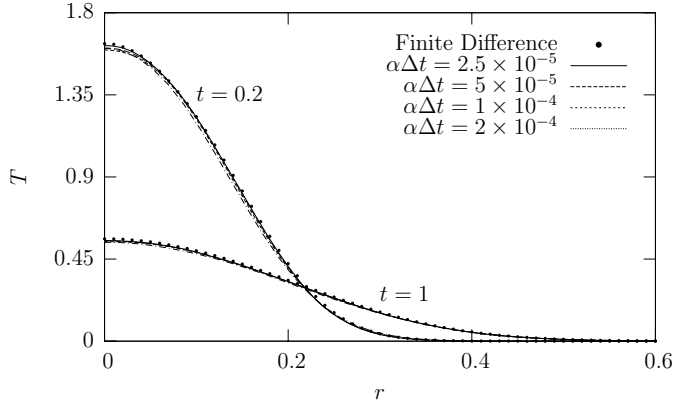


FIGURE 1. NONLINEAR ISOBARIC DIFFUSION OF AN IDEAL GAS WITH $E_0 = 0.2$ ($T_{max} = 5.3$).

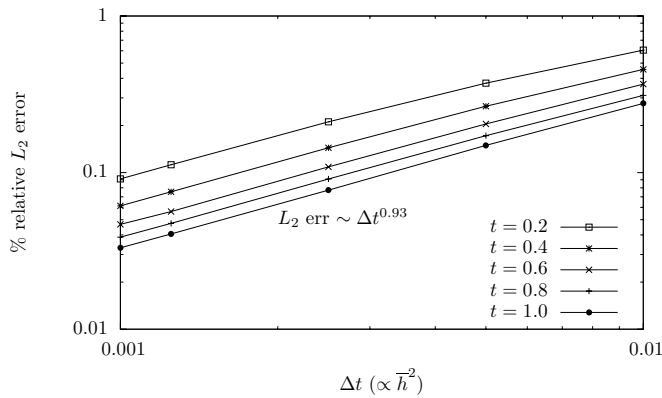


FIGURE 2. L_2 RELATIVE ERROR VERSUS Δt AT DIFFERENT TIMES.

$100 \times \sum |r_i T_i - r_i T_i^*|^2 / \sum |r_i T_i^*|^2$, is plotted in Fig. 2 versus the time step at different times. The error shows approximately linear dependence on Δt and is consequently second order in space, since $\bar{h}^2 \propto \Delta t$. This accuracy is obtained despite the fact that the core function employed is the Green function of the linear diffusion equation with overlap $\sigma/h = 2$ and that moments only up to $m + n = 2$ are conserved.

Natural Convection of a Thermal Patch

We consider low Mach number natural convection of a thermal patch of an ideal gas with an initial temperature distribution is $T_0(\vec{x}) = E_0 / (\pi \sigma_0^2) e^{-|\vec{x}|^2 / \sigma_0^2}$ with $\sigma_0 = 0.25$, $E_0 = T_{0,max} \pi \sigma_0^2$, where $T_{0,max}$ is the maximum initial

temperature. For this problem, all the components contributing to vorticity generation are included. Nonlinear diffusion of vorticity and temperature is computed according the Smooth Redistribution Method using the Maximum Entropy Gradient Projection Algorithm described in section 4. Vorticity generation at elements locations due to baroclinic effects and volumetric expansion is converted to changes in elements circulations using the constrained APRIM(GMRES-RES) Method discussed in section 5. In addition to the velocity component due to vorticity, the velocity component due to volumetric expansion is also included.

In all the simulations, Prandtl number is set to one ($\alpha = \nu$ and $\sigma = \sigma_E = 3h$). Simulations are presented for two sets of cases. The first set consists of three values of Grashof number $Gr = \frac{g T_{0,max} (2\sigma_0)^3}{\nu^2} = 1562.5, 6250$ and 25000 corresponding to $\nu = 0.02, 0.01$ and 0.005, with $T_{0,max} = 0.5$. The second set consists of three values of Grashof number $Gr = 6250, 12500$ and 25000 corresponding to $T_{0,max} = 0.5, 1$ and 2, with $\nu = 0.01$.

Vorticity and Temperature Distributions. Vorticity and temperature contours at $t = 1, 2, 3, 4$ are shown in Fig. 3 for the first set of cases with Grashof number $Gr = 1562.5, 6250$ and 25000 corresponding to $\nu = 0.02, 0.01$ and 0.005 respectively, with $T_{0,max} = 0.5$. The time step is $\Delta t = 0.005$ and the overlap ratio is $\sigma/h = 3$. It can be observed, that except for the role of diffusion in spreading out temperature and vorticity, the three cases are similar. Initially, vorticity is generated at a large rate due to the localized gradients of the temperature distribution. The two vortices pull each other and speed upwards. As time progresses, thermal diffusion weakens the temperature gradients thus generating less vorticity. This is further illustrated in Fig. 4, which shows the time evolution of the total positive generated circulation, where the cases with smaller viscosity experience slight oscillatory behavior. This behavior is more pronounced in the speed of the mean vertical thermal position, given by $d(\int y T dV / \int T dV) / dt$, plotted against time in Fig. 5. The flows in all of the cases considered experience the same steady rise from $t = 0$ to $t \sim 0.3$ after which they experience an oscillatory decelerating behavior. It is further observed that as the viscosity decreases, both amplitude and frequency of these oscillations increase. It can also be observed from Fig. 3 that the rolling vor-

tices entrain cold air creating reverse temperature gradients which introduce pockets of vorticity above the main vortices of opposite sign.

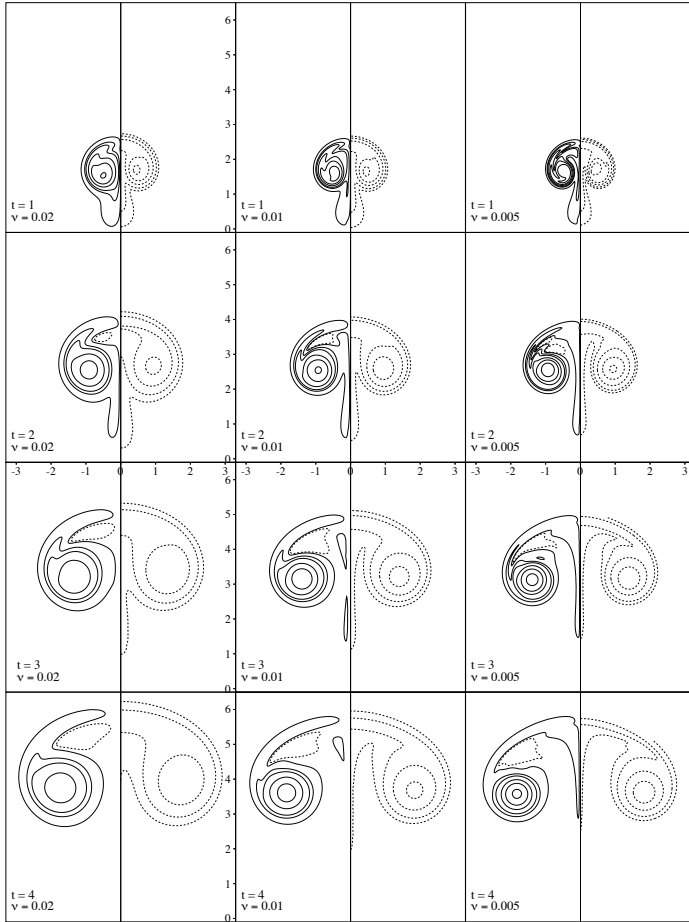


FIGURE 3. VORTICITY (LEFT HALF) AND TEMPERATURE (RIGHT HALF) CONTOURS FOR THE CASES $\nu = 0.02, 0.01$, AND 0.005 . TEMPERATURE CONTOUR VALUES (INWARDS): $0.0001, 0.001, 0.01, 0.5, 0.15$. VORTICITY CONTOUR VALUES (INWARDS): POSITIVE (SOLID): $0.1, 1, 2, 5, 10$ AND NEGATIVE (DASHED): $-0.1, -1, -2, -5, -10$. $T_{0,max} = 0.5$ AND $\sigma/h = 3$.

Results for the second set of cases are presented in Figs. 6, 7 and 8. The total positive circulation is plotted against time in Fig. 6 for the cases $Gr = 6250, 12500$ and 25000 with $T_{0,max} = 0.5, 1$ and 2 respectively, with $\nu = 0.01$. As expected, more vorticity is generated for larger $T_{0,max}$ due to the larger temperature gradient. For

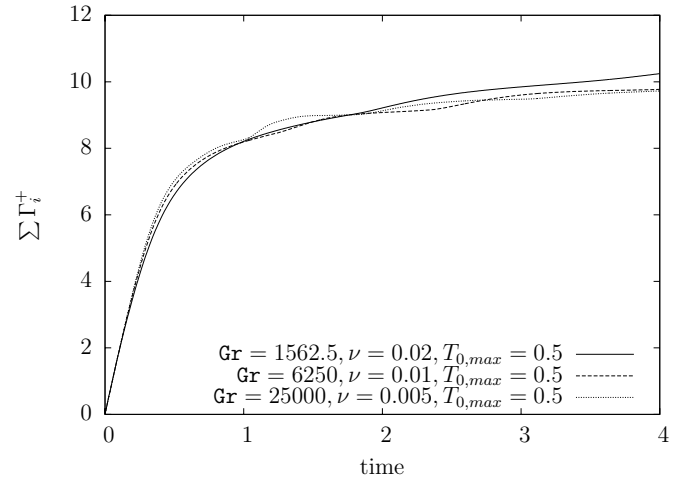


FIGURE 4. TOTAL POSITIVE CIRCULATION VERSUS TIME FOR THE CASES $\nu = 0.02, 0.01$, AND 0.005 , $T_{0,max} = 0.5$ AND $\sigma/h = 3$.

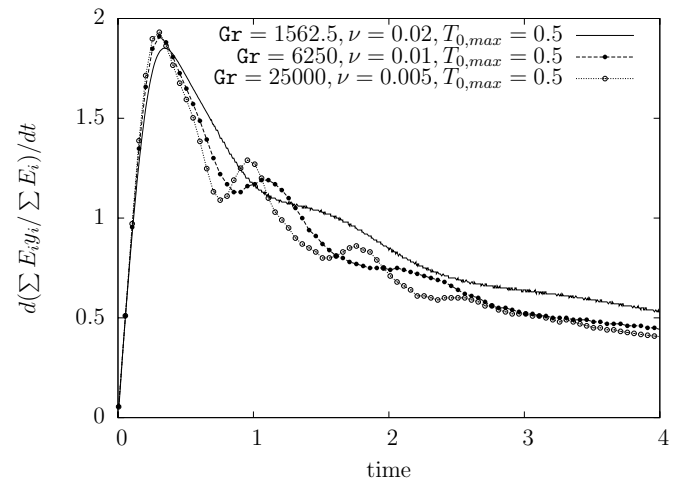


FIGURE 5. SPEED OF THE MEAN THERMAL POSITION VERSUS TIME FOR THE CASES $\nu = 0.02, 0.01$, AND 0.005 , $T_{0,max} = 0.5$ AND $\sigma/h = 3$.

example, at $t = 3$, the total positive circulation is $9.6, 14.3, 21.38$ for $T_{0,max} = 0.5, 1$ and 2 respectively. It can also be seen that for the case $T_{0,max} = 2$, the total positive circulation experiences some drop at $\sim t = 0.6$, but then increases again at $\sim t = 0.7$. This may be attributed to the combined interaction of entrainment of ambient temperature fluid of negative vorticity (see Fig. 8) and diffusion, which together reduce the positive circulation. Same happens to the total

negative circulation. The speed of the mean thermal position, presented in Fig. 7 versus time, shows that the higher $T_{0,max}$ flows rise at a larger speed due to the larger circulation generated. In the deceleration phase, strong oscillations are observed for larger $T_{0,max}$. This is again attributed to the stronger role of convection at the expense of diffusion in moving the temperature gradient and entraining cold fluid of opposite vorticity. Notice from Fig. 8 that for the case $T_{0,max} = 2$, entrainment starts at $t \sim 0.35$, which is about the same time the oscillating decelerating phase begins.

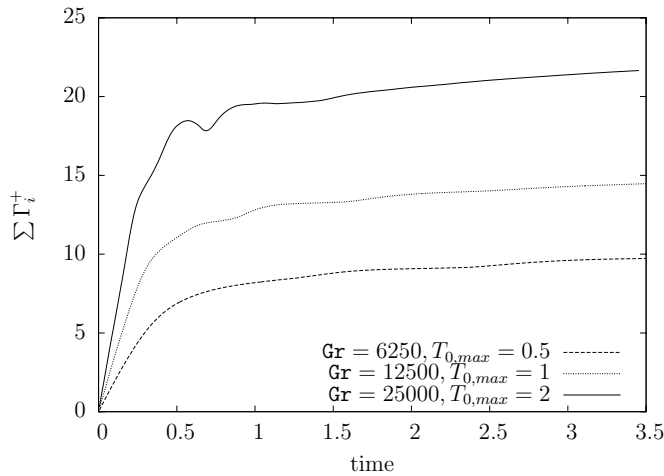


FIGURE 6. TOTAL POSITIVE CIRCULATION VERSUS TIME FOR THE CASES $T_{0,max} = 0.5, 1$, AND 2 , $\nu = 0.01$ AND $\sigma/h = 3$.

CONCLUSION

Grid-free vortex methods were developed for simulation of buoyancy driven ideal gas flows at low Mach number in two-dimensional unbounded domains. The smoothed redistribution method for diffusion was extended to handle nonlinear isobaric diffusion of an ideal gas. The inverse problem relating elements circulations to the vorticity generated due to baroclinic effects and volumetric expansion is solved using the GMRES-based Adaptive Projected Restarted Iterative Method subject to the constraint that the circulation must have the same sign as the vorticity generated at an element's position. Performance of the proposed grid-free vortex method is investigated through simulations of natural convection of a thermal patch for different values of Grashof number.

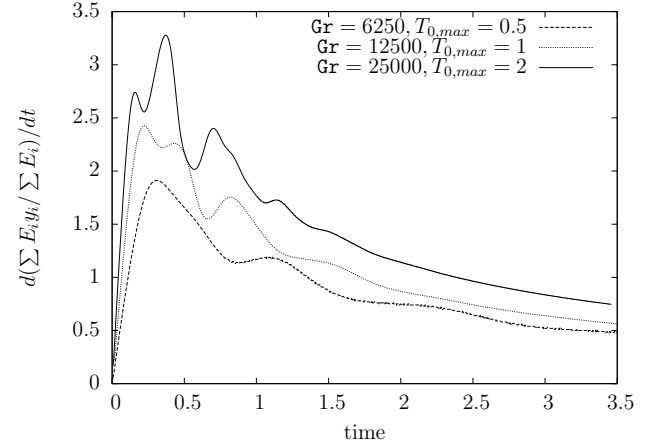


FIGURE 7. SPEED OF THE MEAN THERMAL POSITION VERSUS TIME FOR THE CASES $T_{0,max} = 0.5, 1$, AND 2 , $\nu = 0.01$ AND $\sigma/h = 3$.

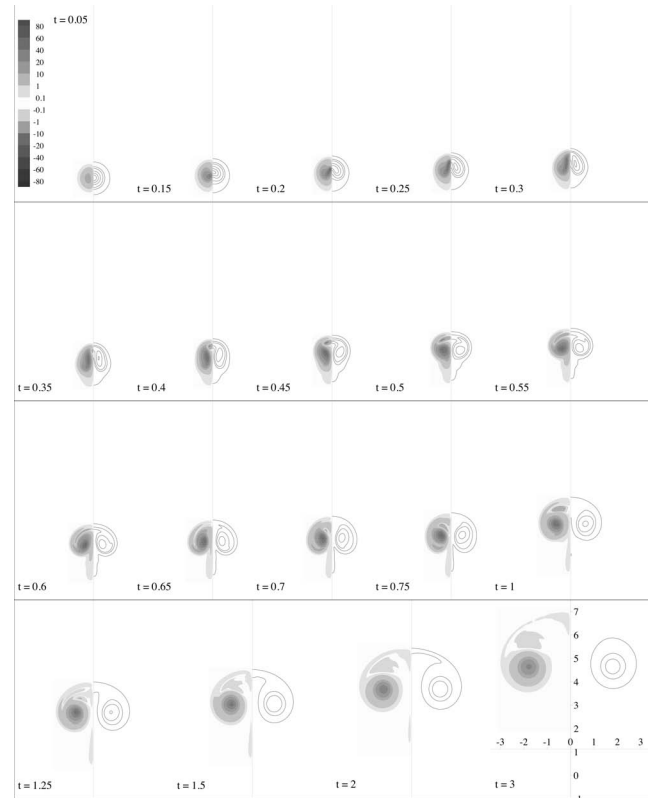


FIGURE 8. VORTICITY (COLORED ON LEFT) AND TEMPERATURE CONTOURS (SOLID LINES ON RIGHT) FOR THE CASE $Gr = 25000$ with $T_{0,max} = 2$, $\nu = 0.01$ AND $\sigma/h = 3$.

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